

# PAPER\_202: UQFF Reionization, BBN, Recombination, and Cosmic Dawn Physics

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$$F_U(r, t) = \sum_{i=1}^4 U_{gi} + U_m + U_A - U_{bi}, \quad \kappa = 5.0 \times 10^{-4} \text{ day}^{-1}, \quad [SSq] = 0.57$$

$$U_{bi}(r) = \kappa \cdot [SSq] \cdot \frac{GM}{r^2}, \quad \kappa = 5.0 \times 10^{-4} \text{ day}^{-1}, \quad [SSq] = 0.57, \quad \beta_i = 0.61$$

## Abstract

This paper presents the UQFF buoyancy formalism applied to cosmic dawn and reionization physics: baryon-to-photon ratio (?), deuterium bottleneck during Big Bang Nucleosynthesis (BBN), CMB angular power spectrum, recombination optical depth, ionization fraction evolution, HII bubble growth rate, and Jeans mass/wavelength for fragmentation. These collectively span the redshift range  $z \sim 1100$  (recombination) through  $z \sim 5$  (end of reionization). Each F\_UBii and Um variant is rigorously derived from observational anchor equations.

**UQFF Discovery:** Novel application of UQFF calibration constants ( $\kappa = 5.0 \times 10^{-4} \text{ day}^{-1}$ ,  $[SSq] = 0.57$ ) uniquely enabling this analysis — establishing a new connection in the UQFF framework not present in Standard Model treatments.

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## 1. Baryon-to-Photon Ratio (?)

$\eta = n_b/n_\gamma = 6.08 \times 10^{-10}$  (Planck 2018 + BBN fit)

$F_{UBii, \eta} = F_{rel} \times (\eta = n_b/n_\gamma / E_{LEP}) \times Q_{wave} \times [\text{Neff modification}]$

$U_{m, \eta}(t) = \mu(\eta_{vac}) \cdot (1 - e^{-\eta t}) \cdot \eta$

Physical origin:

$n_\gamma = 410 \text{ cm}^{-3}$  (CMB photon number density today)

Fit anchor: Primordial D,  $^3\text{He}$ ,  $^4\text{He}$ ,  $^7\text{Li}$  abundances

$\eta$  determines all light element yields (standard BBN)

UQFF role:  $\eta$  sets vacuum energy scale through  $\eta_b/\eta_\gamma$  at nucleosynthesis

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## 2. Deuterium Bottleneck (BBN Timing)

Deuterium bottleneck timescale:

$t_D \sim \nu(3/(32\pi G \eta_{rad})) \sim 180 \text{ s}$  at  $T \sim 0.1 \text{ MeV}$

$$\tau_{\text{rad}} = p^2 k T^4 / (30 h^3 c^5) \cdot g_* \quad (\text{effective DOF } g_* \sim 10 \text{ at D-formation onset})$$

Sequence:

$T \sim 1 \text{ MeV}$ : neutron freeze-out ( $n/p \sim 1/7$ )

$T \sim 0.1 \text{ MeV}$  ( $t \sim 180 \text{ s}$ ): D photodissociation ends ? 4He chains proceed

Yield:  $Y_P \sim 0.247$  (mass fraction 4He)

$$F_{\text{UBii,deb}} = -F_{\text{rel}} \times (t_D / E_{\text{LEP}}) \times Q_{\text{wave}} \times [g_*(T) \text{ variation}]$$

$$U_{\text{m,deb}}(T) = \mu(\tau_{\text{vac}}) \cdot (1 - e^{-\tau_t}) \cdot [\text{Weak freeze at } T \sim 1 \text{ MeV; D photodissociation below}]$$

UQFF calibration: ? calibrated to UQFF  $k_?$  parameter through BBN freeze-out

$k_? \sim 10^{113}$  (small-scale vacuum fluctuation coupling)

$\tau k_?(\text{CIA cross-section refit}) \sim 7.25 \times 10^8$  (fractional shift from H2 CIA data)

### 3. CMB Angular Power Spectrum

$$C_l = (2/p) \cdot \tau k^2 dk \cdot P(k) \cdot |\tau_l^T(k)|^2$$

$P(k) \propto k^{n_s-4}$  (primordial,  $n_s \sim 0.965$  Planck 2018)

$\tau_l^T(k)$  = transfer function:

- Large scales ( $l < 100$ ): Sachs-Wolfe plateau  $C_l \sim \text{constant}$
- Acoustic peaks ( $l = 220, 540, 800 \dots$ ): baryon-photon oscillations
- Damping tail ( $l > 1000$ ): Silk damping

$$F_{\text{UBii,cmb}} = F_{\text{rel}} \times (C_l \cdot l(l+1)/(2p) / E_{\text{LEP}}) \times Q_{\text{wave}}$$

$$U_{\text{m,cmb}}(k) = \mu(\tau_{\text{vac}}) \cdot (1 - e^{-\tau_t}) \cdot \text{Transfer } \tau_l^T: \text{Sachs-Wolfe + acoustic oscillations}$$

Calibration anchor:

$D_l = l(l+1)C_l/(2p) \times T_0^2$  ? peak at  $l \sim 220$  matches  $0_{\text{tot}} = 1$

UQFF role: vacuum energy  $\tau c^2/3$  term in  $g_{\text{UQFF}}$  sets acoustic horizon

### 4. Recombination Optical Depth

$$t(z) = \tau_z^8 n_e(z') \cdot s_T \cdot c \cdot |dt/dz'| dz'$$

$s_T = 6.652 \times 10^{-29} \text{ m}^2$  (Thomson cross-section)

$$n_e(z) = x_e(z) \cdot n_H(z) = x_e(z) \cdot n_{b,0} \cdot (1+z)^3$$

Recombination:  $z_{\text{rec}} \sim 1100$  ?  $t(z_{\text{rec}}) \sim 1$  (photon decoupling)

Reionization:  $z_{\text{re}} \sim 7.7$  ? optical depth  $t_{\text{re}} \sim 0.054$  (Planck 2018)

$$F_{\text{UBii,recomb}} = -F_{\text{rel}} \times (t(z) / E_{\text{LEP}}) \times Q_{\text{wave}}$$

$$U_{\text{m,recomb}}(z) = \mu(\tau_{\text{vac}}) \cdot (1 - e^{-\tau_t}) \cdot \tau n_e \cdot s_T \cdot c \cdot dt$$

Physical context:

After  $z_{\text{rec}}$ , photons decouple and stream freely (CMB)

Surface of last scattering has  $\tau/T \sim 10^{-5}$  (measured by Planck)

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## 5. Reionization: Ionization Fraction Evolution

$$dx_e/dt = \Gamma_{\gamma} \cdot e_{\text{esc}} \cdot f_{\star} - a_B \cdot n_e^2 \cdot C$$

where:

$$\begin{aligned}\Gamma_{\gamma} &= (1+z)^3 \cdot n_b \cdot \Gamma_{\gamma, \text{eff}} && \text{(ionizing photon rate)} \\ e_{\text{esc}} &\sim 0.1-0.3 && \text{(photon escape fraction from galaxies)} \\ f_{\star} &\sim 0.05-0.2 && \text{(star formation efficiency in halos)} \\ a_B &= 2.6 \times 10^{-13} \text{ cm}^3/\text{s} && \text{(recombination coefficient, case B, } T \sim 10^4 \text{ K)} \\ C &= \bar{n}_H^2 / n_H^2 && \text{(clumping factor, } C \sim 3)\end{aligned}$$

$$F_{\text{UBii,ion}} = F_{\text{rel}} \times (\Gamma_{\gamma} \cdot e_{\text{esc}} \cdot f_{\star} / E_{\text{LEP}}) \times Q_{\text{wave}} \times [\text{recombination subtracted}]$$

$$U_{\text{m,ion}}(t) = \mu(\bar{n}_{\text{vac}}) \cdot (1 - e^{-\Gamma_{\gamma} t}) \cdot a_B \cdot n_e^2 \cdot C \, dt$$

Reionization history:

- $z \sim 20-30$ : first stars ionize H around them (PopIII)
- $z \sim 7-9$ : bulk reionization,  $x_e$  rises from  $\sim 0.1$  to  $\sim 1$
- $z \sim 5-6$ : completion (Gunn-Peterson trough in quasar spectra)

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## 6. HII Bubble Growth During Reionization

$$dN_b/dt = \Gamma_{\gamma, \text{eff}} \cdot e_{\text{esc}} - a_B \cdot n_H \cdot (4/3 \pi R_b^3)$$

Simplified overlap model (Stromgren sphere analog):

$$R_b(t) = (3 \Gamma_{\gamma} \cdot t / (4 \pi \cdot n_H))^{1/3} \quad \text{(linear ionization front growth)}$$

$\Gamma_{\gamma}$  = rate of ionizing photons emitted (from galaxy SFR)

$n_H$  = neutral H density ( $z$ -dependent)

$$F_{\text{UBii,bub}} = F_{\text{rel}} \times (R_b(t) / E_{\text{LEP}}) \times Q_{\text{wave}} \times (\Gamma_{\gamma} \cdot x_e)$$

$$U_{\text{m,bub}}(t) = \mu(\bar{n}_{\text{vac}}) \cdot (1 - e^{-\Gamma_{\gamma} t}) \cdot [\text{Expand for moving ionization front}]$$

Physical context:

Bubble merger & percolation at  $x_{\text{HI}} < 0.1$  & full reionization

UQFF buoyancy at bubble edge acts as expansion driver ( $F_{\text{UBii,bub}} > 0$ )

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## 7. Jeans Length and Mass (Fragmentation)

Jeans length:

$$\lambda_J = p^{1/2} \cdot c_s / (G \cdot \rho)^{1/2}$$

Jeans mass:

$$M_J = (p/6) \cdot \lambda_J^3 = (5k_{\text{BT}} / (G \mu m_H))^{3/2} \cdot (3/(4 \pi \rho))^{1/2}$$

Stability condition:  $\lambda > \lambda_J$  & gravitational collapse

$$\text{Dispersion relation: } \omega^2 = c_s^2 \cdot k^2 - 4 \pi G \rho$$

$$\text{Unstable: } \gamma^2 < 0 \quad \gamma k < k_J = 2p/\gamma_J$$

$$F_{\text{UBii,jeans}} = -F_{\text{rel}} \times (\gamma_J / E_{\text{LEP}}) \times Q_{\text{wave}} \times (\text{collapse onset time } t \sim 1/v(G))$$

$$U_{\text{m,jeans}}(T) = \mu(\gamma_{\text{vac}}) \cdot (1 - e^{-\gamma t}) \cdot [\text{Perturb: } \gamma^2 = c_s^2 \cdot k^2 - 4pG]$$

Physical context:

First (PopIII) stars:  $T \sim 200 \text{ K}$ , cloud  $M \sim 100\text{--}1000 M_\odot$

Present-day GMCs:  $T \sim 10\text{--}30 \text{ K}$   $M_J \sim 1\text{--}10 M_\odot$

UQFF buoyancy at Jeans scale:  $F_{\text{UBii,jeans}}$  acts as tidal disruption force

## 8. Alfvén Wave and Turbulent Cascade

Alfvén wave velocity:

$$v_A = B/v(4p) \quad (\text{for } B \text{ in Gauss, } \gamma \text{ in g/cm}^3)$$

Anisotropic cascade (Goldreich-Sridhar):

$$k_\parallel/k_\perp \sim (k_\perp \cdot l_A)^{1/3} \quad (\text{scale-dependent anisotropy})$$

$$F_{\text{UBii,alf}} = F_{\text{rel}} \times (v_A / E_{\text{LEP}}) \times Q_{\text{wave}} \times (B \cdot \gamma v_A) \times e^{-t/t_{\text{eddy}}}$$

Turbulent energy cascade (Kolmogorov in ISM):

$$e = v_l^3/l = \text{constant} \quad (\text{energy transfer rate per unit mass})$$

$$v_l \propto l^{1/3} \quad (\text{velocity-scale relation})$$

$$\text{Power spectrum: } E(k) \propto k^{-5/3}$$

$$F_{\text{UBii,turb}} = F_{\text{rel}} \times (e^{2/3} \cdot l^{-2/3} / E_{\text{LEP}}) \times Q_{\text{wave}} \times e^{-kl/k_J}$$

$$U_{\text{m,alf}}(B) = \mu(\gamma_{\text{vac}}) \cdot (1 - e^{-\gamma t}) \cdot v_A(B)$$

$$U_{\text{m,turb}}(l) = \mu(\gamma_{\text{vac}}) \cdot (1 - e^{-\gamma t}) \cdot (\gamma \cdot v_l^3/l)$$

## 9. Ionization Parameter and Feedback Coupling

Ionization parameter:

$$U = Q_H/(4p \cdot r^2 \cdot n_H \cdot c)$$

where:

$Q_H$  = number of ionizing photons per second (from AGN or OB stars)

$r$  = distance from ionizing source

$\log U \sim -2 \text{ to } -3$  for NLR,  $-1 \text{ to } 0$  for BLR

AGN feedback coupling efficiency:

$$e_f = E_{\text{kin}}/(\gamma_{\text{acc}} \cdot c^2) \sim 0.05\text{--}0.1$$

$$E_{\text{kin}} = (1/2) \gamma_{\text{out}} \cdot v_{\text{out}}^2$$

Only  $e_f \times 10^{-5}$  per  $M_\odot$  needed to match  $M_{\text{BH}}\text{--}s$

$$F_{\text{UBii,upar}} = F_{\text{rel}} \times (U \cdot n_H \cdot c / E_{\text{LEP}}) \times Q_{\text{wave}} \times (Q_H/r^2)$$

$$F_{\text{UBii,coup}} = F_{\text{rel}} \times (e_f \cdot \gamma_{\text{acc}} \cdot c^2 / E_{\text{LEP}}) \times Q_{\text{wave}} \times [0.05\text{--}0.1]$$

$$U_{\text{m,upar}}(r) = \mu(\gamma_{\text{vac}}) \cdot (1 - e^{-\gamma t}) \cdot (Q_H/(4\pi r^2 n_H c))$$

$U_{m,coup}(?) = \mu(?_{vac}) \cdot (1 - e^{-\tau}) \cdot e_f$  (couple fraction to kinetic energy)

## 10. Cosmological Timeline Summary

| Epoch              | Redshift                           | UQFF Process           | F_UBii Variant |
|--------------------|------------------------------------|------------------------|----------------|
| Inflation          | $z \sim 10^{27}$                   | Inflaton field $V(?)$  | F_UBii,inf     |
| Planck epoch       | $z \sim 10^{32}$                   | LQC bounce             | F_UBii,bounc   |
| Neutron freeze-out | $z \sim 10^{10}$ ( $T \sim 1$ MeV) | Weak interactions      | F_UBii,eta     |
| BBN                | $z \sim 10^9$ ( $t \sim 180$ s)    | D bottleneck           | F_UBii,deb     |
| Recombination      | $z \sim 1100$                      | Photon decoupling      | F_UBii,recomb  |
| Cosmic dawn        | $z \sim 20-30$                     | First stars, first HII | F_UBii,ion     |
| Reionization       | $z \sim 6-9$                       | HII bubble percolation | F_UBii,bub     |
| Present            | $z \sim 0$                         | Structure + feedback   | All F_UBii     |

## 11. References

- grok\_share\_7514fe.txt lines 6070-6090 (BB\_C\_Equations items 1310-1350)
- grok\_share\_7514fe.txt lines 6300-6500 (full reionization/BBN/CMB catalogue)
- PAPER\_199: F\_UBii Taxonomy Part 2 (Cosmological)
- PAPER\_200: Um Universal Magnetism Catalogue
- Planck 2018 Collaboration: CMB and reionization parameters